Street Food Festival back end

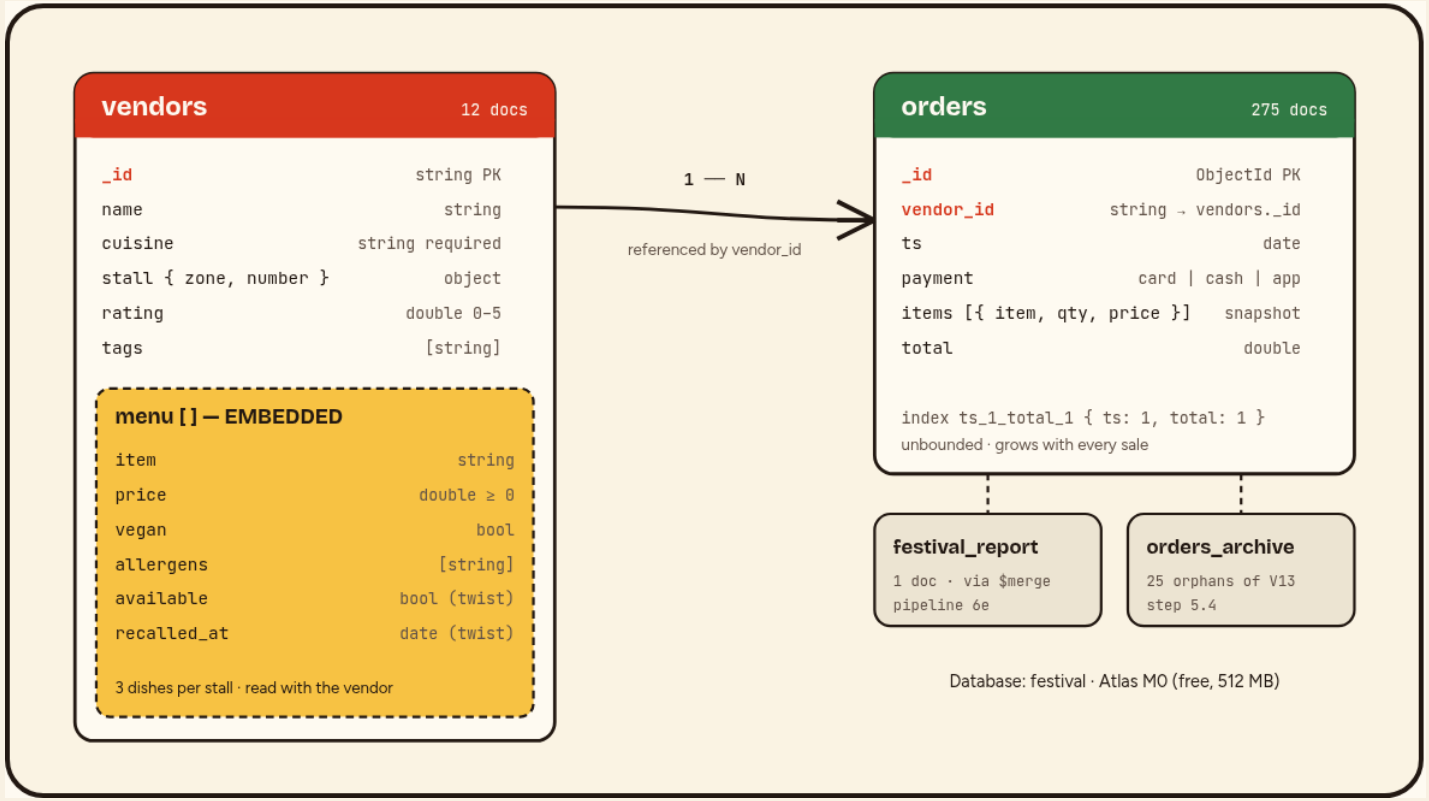
Which stalls earn most · when queues peak · how fast we warn about an allergen

12	9	300	5	275 → 0
VENDORS	CUISINES	ORDERS	PIPELINES	DOCS EXAMINED

Team: [Member 1] · [Member 2] · [Member 3]

Free Atlas M0 cluster · database festival · collections vendors & orders

Embed the menu, reference the orders



Menu embedded: small, bounded, always read with the vendor, rarely changes → one read gives every price and allergen.

Orders referenced by vendor_id: unbounded, would grow vendors toward 16 MB and rewrite them on every sale.

Orders can be indexed by time and aggregated across all vendors; each keeps a price snapshot.

Size: order ≈ 150 B → ~3 M orders fit in 512 MB.

What the organisers learn

6a Revenue per vendor

Sushi Wave 448.50 €

Napoli Slice 389.00 €

Tbilisi Fire 379.00 €

Total 3 782 € · 275 orders

6b Busiest hour per day

20 Jun 12:00–13:00 (11)

21 Jun 15:00–16:00 (13)

22 Jun 14:00–15:00 (16)

→ staff up at lunch

6c Best dish per cuisine

Mexican: Chicken quesadilla 26

Georgian: Khachapuri 25

Ukrainian: Varenyky 22

Japanese: Avocado maki 21

6d Payment share

Card 35.3 %

App 32.4 %

Cash 32.4 %

6e report → $merge

`$group` · `$sort` · `$hour` · `$dayOfMonth` · `$unwind` · `$lookup` · `$first` · `$facet` · `$merge`

Index { ts: 1, total: 1 } → covered query

explain("executionStats") on pipeline 6b

275

docs before · COLLSCAN

0

docs after · IXSCAN covered

ts carries the range filter; total is added so the index holds every field the pipeline reads — no document is opened.

```
$jsonSchema on vendors
required: [_id, name, cuisine,
           stall, menu]
menu.items.price: { minimum: 0 }

insertOne({ ... price: -2.5 })
→ REJECTED: Document failed validation
  itemIndex 0, minimum 0

insertOne({ no cuisine })
→ REJECTED: missingProperties
  ['cuisine']
```

One updateMany, nine safe stalls

```
db.vendors.updateMany(
  { "menu.allergens": "sesame" },
  { $set: { "menu.$[dish].available": false } },
  { arrayFilters: [{ "dish.allergens": "sesame" }] })
// matched 3 · modified 3 · 7 dishes recalled

db.vendors.find({ "menu.allergens": { $ne: "sesame" } })
```

Safe for sesame allergy

Taco Loco A1 · Napoli Slice A2

Pelmeņi Express A3 · Rīgas Kūpinātava A5

Tbilisi Fire B1 · Bombay Street B3

Supra Grill B4 · Borsch & Co C1 · El Fuego C5

Avoid: Tokyo Ramen Bar, Sushi Wave, Beirut Bites (cross-contamination risk).

Bonus: 11 recalled portions sold after 19:00 → visitors to contact. Deleting a vendor leaves orphan orders → we archive them.